

CROSS-LAYER SEEDED FAULT INJECTION ARCHITECTURE

Systematic fault injection across Sensory, Compute/Bus, Model/VLA, and Electrical layers into the Dual-Brain Platform

1. SENSORY & TIMING FAULTS	2. COMPUTE & BUS FAULTS	3. MODEL & PROPOSAL FAULTS	4. ELECTRICAL & PHYSICAL
• PTP Timestamp Drift: +35 ms phase skew. • Camera Occlusion: 500 ms black frames. • MIPI CRC Corruption: 5% dropped packets. • IMU Bias Jump: Sudden +0.5 rad/s offset.	• Linux Kernel Panic: MPU hard freeze. • UMA Memory Bus Stall: 95% saturation. • Python GIL Lock: 150 ms GC pause. • DVFS Thermal Throttle: 1.8 GHz → 600 MHz.	• Out-of-Distribution Prompt: Nonsense string. • Hallucinated Grasp Pose: Target behind table. • Stale Intent Lease: Emitted 800 ms late. • Singular Trajectory: High condition number.	• Battery Voltage Sag: 24V → 18V transient. • Motor Overheating: Coil temp > 110°C. • Dynamic Obstacle: Sudden trajectory block. • Safety Interlock Trip: E-stop pushed.

✓ REAL-TIME MCU CONTAINMENT VERDICT: 100% CONTAINMENT OF ALL INJECTED VECTORS

1. HARDWARE WATCHDOG INTERLOCK

- Detects MPU stall in < 50 ms.
- Instantly trips Category 0 STO.
- Zero software cooperation required.

2. CBF BARRIER PROJECTION

- Bounds trajectory in 65 μs.
- Prevents obstacle collision.
- Seamlessly projects into safe set.

3. EXPIRING LEASE REVOCATION

- Rejects stale proposals (85mm drift).
- Executes Category 2 Position Hold.
- Flushes MPU action queue buffers.

Release Requirement: Zero physical collisions or current overstress across 5,000 automated bench fault injection runs.